

Biomimetics

Biomimetics – from the Greek meaning ‘imitation of life’ – describes the study of biological structures, processes and behaviours, and its application in various fields, such as product design or structural engineering. Observing how natural organisms adapt to their environments can help create efficient and/or sustainable designs (on the premise that if nature has honed a design over millions of years, it must be good). Natural structures have long inspired architects; for instance, the fan vaulting in church naves (from medieval times to Gaudí’s Sagrada Família, opposite) mimics the branching of tree canopies. Other examples that borrow structurally from nature include the Eden Project (2001) in Cornwall, modelled after soap bubbles and pollen grains, and Foster’s Gherkin Tower (2003) in London, with a hexagonal skin inspired by a marine sponge. The Eastgate Centre in Harare (1996), by local architect Mick Pearce (b.1938), was designed to mimic the natural thermal control of a termite mound, as an alternative to using air-conditioning systems.

